

Team Profile

2026 Global Quantum + AI Challenge, HSBC problem statement

September 2026

Claude Shannon’s Fraud Catchers

Problem statement: HSBC, Quantum-Enhanced Credit Card Fraud Detection for Digital Payment Ecosystems.

Team lead and sole participant. Harold I. Kimmey III (“Harold Kimmey”), independent researcher (unaffiliated). Role: principal investigator, holding all decision authority for this submission. Email: kimmeyharold@aol.com. Phone: +1 216-357-9227. LinkedIn: <https://linkedin.com/in/haroldkimmey>

Composition and disclosure. A solo entry with no other members. Harold Kimmey is the sole legal participant and is responsible for every claim in the submission. Claude Code (Anthropic) was used throughout as a development tool under the team lead’s direction; every experimental decision, protocol change, and metered hardware execution required his explicit approval, recorded in the amendment log and issue history. Large language models were also used as adversarial reviewers of the submission documents – OpenAI’s ChatGPT and Anthropic’s Claude, prompted to find errors, unsupported claims and internal contradictions rather than to write content. Several defects recorded in the amendment log were found that way, including three that changed published figures. Every finding was verified against the code and artifacts before acceptance, and some were rejected as incorrect. The team name references Claude Shannon and Claude Code, and does not imply OpenAI’s or Anthropic’s participation or endorsement.

What this entry delivers. Four artifacts, all produced within Phase 1 and all inspectable: a frozen preregistration with thirty-two dated amendments, under which every gate was scored as committed, including the one that failed and whose stopping rule we did not honour; a completed hardware campaign of 61 metered Dirac-3 fits over 1,141 metered device seconds with zero device failures, 48 raw responses version-controlled and job identifiers for 60 of the 61 fits; a reproducibility package with analysis code, a results store carrying a configuration hash on every row, a dataset checksum manifest, and 322 tests including known-answer checks on the controls; and a production-trial design specifying what a bank would run in 90 days and what it would have to satisfy to proceed.

Relevant experience. Prior campaigns on QCI’s Dirac-3 using the continuous-variable boosting family, including the tuned configuration that seeds this submission’s quantum arm. Quantum machine learning breadth across reservoir computing, phase-based feature engineering, and explainability methods. Hands-on Amazon Braket experience within a broader AWS background, alongside tutorial work and reproductions of published experiments in Qiskit, PennyLane and TensorFlow Quantum, which is why a Phase 2 gate-based arm is a commitment. Applied machine learning experience on highly imbalanced classification problems and on fraud detection.

Working method and constraints. The submission was produced under a protocol frozen before any result was observed, with dated amendments for every change and evidence tags separating measured hardware results from simulation and projection. A solo entry has limited throughput and the Phase 1 hardware budget was small, which shaped the experiment grid; the remaining blocks are specified and costed rather than claimed.